

Multi-Scale Fusion versus Encoder Capacity for Efficient Visual Saliency Prediction on SALICON

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Abstract—Visual saliency prediction estimates where observers are likely to look in a natural image. This work examines whether multi-scale feature fusion can improve fixation prediction while retaining the efficiency of a lightweight encoder. Four methods are compared on a held-out 2,500-image internal split of SALICON at a resolution of 169×256 : a non-learned MeanMap baseline, a MobileNetV2 single-scale model, a MobileNetV2 multi-scale model, and a ResNet-18 multi-scale reference. All neural models share the same preprocessing, objective, training policy, checkpoint criterion, and evaluation code. The MobileNetV2 multi-scale model achieves a KLD of 0.2257, a CC of 0.8813, and a SIM of 0.7733, improving all three metrics over its single-scale counterpart while adding only 0.27% parameters and 1.20% MACs. The ResNet-18 model requires 6.12 times more parameters and 2.50 times the median CPU latency, yet provides no improvement in KLD or CC under the same protocol. A post-hoc 625-image off-centre subset, selected according to target centre-of-mass displacement, further shows that the learned models remain substantially stronger than the fixed MeanMap prior when attention is displaced from the image centre. Overall, the results suggest that compact multi-scale fusion provides a more favourable accuracy–efficiency trade-off than increasing encoder capacity for the selected split, resolution, and training budget.

Index Terms—visual saliency prediction, fixation prediction, convolutional neural networks, multi-scale feature fusion, efficient deep learning, centre bias

I. INTRODUCTION

Human observers do not inspect every part of a visual scene uniformly. Instead, their gaze tends to concentrate on a limited number of informative regions, such as faces, people, text, and visually distinctive objects. Visual saliency prediction models this behaviour by mapping an RGB image to a dense distribution over likely fixation locations. Unlike salient-object segmentation, the objective is not to produce a binary object mask, but to estimate the graded spatial distribution of visual attention. Such predictions can support applications including adaptive image compression, automatic cropping, streaming, and other systems that allocate resources according to visual importance.

Convolutional encoders naturally produce visual representations at different levels. Deep features contain semantic and global information because of their larger receptive fields, but repeated downsampling reduces spatial detail. Shallower features retain positions, edges, and local contrast, although they contain less semantic context. Previous visual-attention models have shown that combining information from multiple depths can improve fixation prediction [1]. However, multi-scale designs are often associated with large backbones, parallel

branches, or computationally expensive decoders. This raises a practical question: can multi-scale information improve a lightweight saliency model without removing its efficiency advantage?

A second difficulty is the strong preference for central image regions found in many natural-image attention datasets. SALICON provides large-scale free-viewing attention annotations on natural MS COCO scenes [2], but its targets also contain a substantial centre bias. Consequently, a fixed central prediction can obtain non-negligible scores without responding to the content of an individual image. Evaluating prediction accuracy alone may therefore give an incomplete picture. Computational cost and behaviour on scenes with displaced target attention should also be considered.

The aim of this project is not to reproduce a large state-of-the-art saliency architecture, but to isolate two practical design choices under a common experimental protocol: whether multi-scale fusion benefits a lightweight encoder, and whether increasing encoder capacity justifies its additional computational cost. Four methods are considered. MeanMap is a non-learned baseline obtained by averaging the normalized training saliency maps. Light-S combines MobileNetV2 with a decoder that uses only its deepest encoder feature. Light-M retains the same encoder but introduces a compact top-down decoder that combines four feature scales through narrow lateral projections and lightweight refinement blocks. Heavy-M replaces MobileNetV2 with ResNet-18 while preserving the same multi-scale decoder structure. All trainable models use the same data split, output resolution, loss, optimisation policy, checkpoint criterion, and evaluation code.

The main contributions of this work are:

- a compact top-down multi-scale decoder designed to recover spatial detail while retaining a lightweight MobileNetV2 encoder;
- a controlled comparison that separates the contribution of multi-scale fusion from the effect of increasing encoder capacity;
- a joint evaluation of prediction quality and computational cost using KLD, CC, SIM, parameter count, MACs, CPU latency, a MeanMap baseline, and a reproducible post-hoc off-centre subset.

Under the selected experimental protocol, Light-M provides the most favourable accuracy–efficiency trade-off. It improves all three prediction metrics over Light-S at nearly unchanged computational cost, whereas Heavy-M is substantially more expensive without a corresponding improvement in KLD or

CC. The off-centre analysis also provides evidence that the neural predictions cannot be explained by the fixed training-set spatial prior alone.

The remainder of the report is organized as follows. Section II reviews previous saliency-prediction, multi-scale, and efficient-network approaches. Sections III–V describe the processing pipeline, dataset, architectures, training objective, and evaluation protocol. Section VI presents the accuracy, efficiency, centre-bias, and qualitative results. Finally, Section VII summarizes the main findings, limitations, and possible extensions.

II. RELATED WORK

Early computational models of visual attention mainly relied on hand-crafted low-level cues. Itti *et al.* combined intensity, colour, and orientation contrasts to identify image regions that differed from their surroundings [3]. Such methods provided an important basis for saliency prediction, but their predefined features were less suitable for scenes in which attention depends on object identity or semantic context. The availability of larger datasets later supported data-driven approaches. In particular, SALICON introduced large-scale free-viewing annotations collected on natural MS COCO images through a mouse-contingent procedure [2]. The dataset also contains a strong centre bias, which motivates the MeanMap baseline and the off-centre analysis used in this work.

Convolutional neural networks improved fixation prediction by learning hierarchical representations directly from images. Deeper layers contain semantic and global information, whereas shallower layers retain finer spatial detail. Wang and Shen combined features from different depths using a VGG-based skip-layer encoder–decoder with multiple supervised saliency outputs [1]. The present work adopts the general idea that features from different depths are complementary, but does not reproduce their complete architecture. Light-M instead uses one narrow top-down decoder, additive feature fusion, and a single final prediction.

Computational efficiency is relevant when saliency maps must be produced under limited memory or processing resources. MobileNetV2 reduces computation through inverted residual blocks and depthwise separable convolutions [4], while ResNet-18 provides a larger residual encoder for comparison [5]. FastSal further showed that lightweight backbones can provide a useful balance between saliency accuracy and inference cost [6]. Following this efficiency-oriented direction, this project compares single- and multi-scale MobileNetV2 models with a ResNet-18 multi-scale reference under the same experimental protocol.

Saliency predictions can also be interpreted as spatial probability distributions. Jetley *et al.* investigated distribution-based objectives for end-to-end fixation prediction [7]. Moreover, different saliency metrics reward different properties of a predicted map and can produce different model rankings [8]. These observations motivate the use of one common distribution-based training objective and the separate reporting

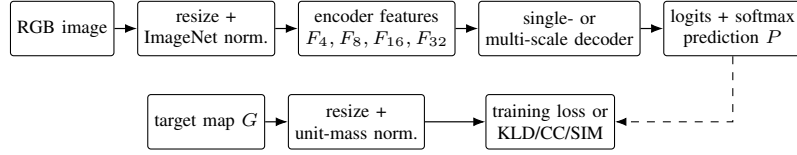


Fig. 1. Common processing pipeline. Light-S uses only F_{32} , while Light-M and Heavy-M fuse features from F_4 to F_{32} .

of KLD, CC, and SIM. Rather than attempting a state-of-the-art benchmark comparison, this work focuses on a controlled study of multi-scale fusion, encoder capacity, computational cost, and sensitivity to the SALICON centre prior.

III. PROCESSING PIPELINE

The project addresses free-viewing fixation prediction. Each sample contains an RGB image and a continuous saliency map showing the distribution of human attention. The same pipeline is used for all neural models, so the comparison mainly focuses on multi-scale fusion and encoder capacity.

Images and maps are paired through their sample identifier, resized to 169×256 pixels, and converted to tensors. Images are normalized with ImageNet statistics, while target maps are made non-negative and normalized to unit mass. An ImageNet-pretrained encoder extracts the features. Light-S uses only the deepest feature, while Light-M and Heavy-M use features from four scales. The decoder produces one-channel logits, which are converted into a spatial probability distribution before computing the loss or the evaluation metrics.

This setup gives two direct comparisons. Light-S and Light-M share MobileNetV2, so they show the effect of multi-scale fusion. Light-M and Heavy-M use the same decoder but different encoders, allowing the effect of encoder capacity to be studied separately.

IV. DATASET AND PREPROCESSING

A. SALICON Data and Splits

The experiments use SALICON, which contains natural MS COCO images and continuous attention maps collected through a mouse-contingent viewing procedure [2]. The maps are treated as spatial distributions rather than binary masks or discrete fixation points.

Images and maps were paired using their identifiers instead of directory order. The dataset audit found 10,000 labelled training pairs and 5,000 labelled official-validation pairs, with no duplicated identifiers. All training pairs are used for learning. The official validation set is shuffled once with seed 42 and divided equally into 2,500 validation samples and 2,500 internal test samples. The hidden official SALICON test annotations are not used.

The same split files are used by all four methods. Validation data are used for checkpoint selection, while the internal test set is evaluated only after the pipeline is fixed. An off-centre subset is later selected from this test set for the centre-bias analysis.

B. Image and Target Preparation

Images are loaded in RGB format and saliency maps as floating-point, one-channel targets. Both are resized to 169×256 using bilinear interpolation and antialiasing. The same resolution is used for training, validation, and testing.

Images are normalized using the ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225). For a target map G , negative values are removed and the remaining mass is normalized as

$$G'_i = \frac{\max(G_i, 0)}{\sum_j \max(G_j, 0) + \epsilon}, \quad (1)$$

where i and j represent spatial positions. No random crop, flip, rotation, or colour augmentation is used in the final training pipeline. All experiments read the same verified files from a temporary local Colab cache, which reduces Drive input/output time without changing the data.

V. LEARNING FRAMEWORK

A. Compared Models

Four methods are evaluated. MeanMap is a non-learned centre-prior baseline. Light-S combines MobileNetV2 with a single-scale decoder. Light-M uses the same encoder with a compact multi-scale decoder, while Heavy-M replaces MobileNetV2 with ResNet-18.

MeanMap is created by normalizing and averaging all training saliency maps. The final average is normalized again and used as the prediction for every image. It measures how much performance can be obtained from the dataset's spatial prior alone.

The neural encoders return features at output strides 4, 8, 16, and 32, written as F_4 , F_8 , F_{16} , and F_{32} . MobileNetV2 provides 24, 32, 96, and 320 channels, while ResNet-18 provides 64, 128, 256, and 512 channels [4], [5]. Both backbones use the same feature interface and decoder logic.

B. Single- and Multi-Scale Decoders

Light-S uses only F_{32} . A 1×1 convolution projects it to 32 channels, after which the feature is upsampled toward strides 16, 8, and 4. Each step applies a lightweight refinement block. A final 1×1 layer produces the saliency logits, which are resized to the target resolution.

Light-M uses all four feature scales. Each feature is first projected to 32 channels. Starting from F_{32} , the decoder up-samples the current representation, adds the projected feature from the next finer scale, and then refines the result:

$$D_k = R_k(\text{Up}(D_{k+1}) + P_k(F_k)), \quad (2)$$

where P_k is the lateral projection, $\text{Up}(\cdot)$ is bilinear interpolation, and R_k is the refinement block. Each block contains a depthwise 3×3 convolution, a pointwise 1×1 convolution, Group Normalization, and ReLU.

The design follows the idea of combining features from different depths [1], but uses one compact top-down decoder and one final prediction instead of multiple supervised branches.

Heavy-M uses ResNet-18 with the same multi-scale decoder structure, width, fusion rule, and refinement blocks. Its lateral projections contain more parameters because the ResNet features have more channels, but the overall decoder design and training protocol remain unchanged.

C. Spatial Output and Objective

Each neural model returns logits z with shape $[B, 1, H, W]$. They are flattened over the spatial dimensions and converted into a probability distribution using spatial softmax:

$$P_i = \frac{\exp(z_i)}{\sum_j \exp(z_j)}. \quad (3)$$

The predicted map is therefore non-negative and sums to one. All models use the same training objective:

$$\mathcal{L} = D_{\text{KL}}(G \| P) + 0.5 (1 - \text{CC}(P, G)). \quad (4)$$

The KLD term compares the predicted and target probability mass, while CC encourages a similar spatial pattern. This distribution-based formulation follows previous saliency-prediction work [7]. SIM is reported during evaluation but is not directly optimized.

D. Training Protocol

The neural models are trained with AdamW and a batch size of 8. The encoder learning rate is 10^{-4} and the decoder learning rate is 3×10^{-4} , with weight decay 10^{-4} . Gradient norms are clipped to 1.0, and mixed precision is used on CUDA.

The encoder is frozen during the first epoch and then fine-tuned while its Batch Normalization running statistics remain fixed. Training lasts at most 15 epochs. The checkpoint with the highest validation CC is selected, with early stopping after three epochs without an improvement of at least 10^{-4} .

E. Evaluation Protocol

Prediction quality is measured using KLD, CC, and SIM. The metrics are kept separate because they measure different properties of a saliency map and may not always give the same model ranking [8].

Efficiency is measured using parameter count, MACs, and model-only CPU latency. MACs are computed for an input of shape $[1, 3, 169, 256]$. CPU latency uses one thread, 30 warm-up forwards, and 200 timed forwards, repeated three times. Data loading and preprocessing are excluded.

Centre-bias dependence is evaluated using MeanMap and an off-centre subset. For each internal-test target, the normalized centre of mass (c_x, c_y) is calculated, together with its distance from the image centre:

$$d = \sqrt{(c_x - 0.5)^2 + (c_y - 0.5)^2}. \quad (5)$$

The 625 images with the largest distance, corresponding to 25% of the test set, form the off-centre subset. This makes it possible to test the models on examples where the central spatial prior is less useful.

VI. RESULTS

The final results are measured on the frozen internal test split of 2,500 images. All methods use the same resolution of 169×256 and the same implementation of Kullback–Leibler divergence (KLD), correlation coefficient (CC), and similarity (SIM). Lower KLD indicates a better result, while higher CC and SIM are better. Efficiency is compared using the number of parameters, multiply–accumulate operations (MACs), and model-only CPU latency.

A. Training Behaviour and Checkpoint Selection

The final checkpoints were selected according to the highest validation CC. Light-S obtained its best value at epoch 5, while Light-M and Heavy-M both reached their best validation CC at epoch 6. After these points, the training loss continued to decrease, but the validation loss started to increase. This effect was clearer for Heavy-M, which had the largest difference between training and validation loss.

This indicates that the models started to overfit slightly after the selected epochs. For this reason, the final evaluation uses the saved `best.pt` checkpoints and not the models from the last epoch. For all three models, the difference between validation CC and test CC was smaller than 0.001. This suggests that the selected checkpoints also behaved consistently on the held-out test split.

B. Overall Accuracy and Efficiency

Table I reports the main results. MeanMap obtains non-zero scores because it represents the average distribution of the training saliency maps. However, its results are clearly worse than those of the learned models. Compared with MeanMap, Light-M reduces KLD by about 68.7% and improves CC by 0.3243. The difference shows that the neural models are using the input image and are not only learning the average centre prior.

Light-M obtains the best KLD and CC. Heavy-M has the highest SIM, but the difference from Light-M is only 0.0004. The three metrics therefore give slightly different rankings. This is not unexpected, since KLD measures the difference in probability mass, CC measures the correlation between the spatial patterns, and SIM measures their overlap. For this reason, the metrics are reported separately rather than being combined into one score.

C. Effect of Multi-Scale Fusion

The comparison between Light-S and Light-M shows the effect of adding multi-scale features. Both models use MobileNetV2, the same dataset split, training loss, optimizer settings, decoder width, and checkpoint rule. The main difference is that Light-S uses only the deepest encoder feature, while Light-M also uses features from shallower stages.

Light-M reduces KLD from 0.2371 to 0.2257, which corresponds to a relative improvement of 4.79%. CC increases from 0.8735 to 0.8813, and SIM increases from 0.7690 to 0.7733. The absolute differences are small, but Light-M improves all three metrics.

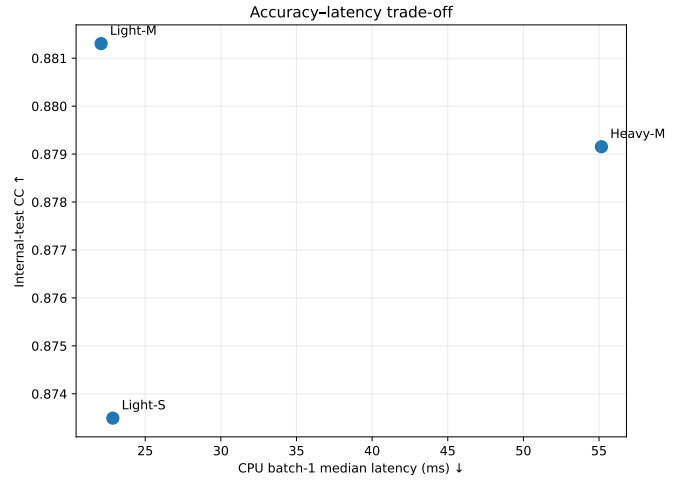


Fig. 2. Accuracy–latency comparison of the neural models. Light-M improves CC over Light-S with similar CPU latency, while Heavy-M is slower without improving CC.

The per-image results also show that the improvement does not come from only a few particular images. Light-M performs better than Light-S on 57.92% of the test images for KLD, 62.32% for CC, and 56.36% for SIM. The clearest improvement is therefore found for spatial correlation.

The extra cost of Light-M is limited. It adds 4,864 parameters, which is only a 0.266% increase over Light-S. Its MAC count increases by approximately 1.20%. The measured CPU latencies are also very similar. Light-M was slightly faster in this measurement, but the difference is smaller than the variation between repeated timing runs. It should therefore be considered as similar latency rather than a real speed improvement.

D. Effect of Encoder Capacity

Light-M and Heavy-M use the same four-scale decoder, but Heavy-M uses ResNet-18 instead of MobileNetV2. This comparison checks whether the larger encoder gives enough accuracy improvement to justify its higher cost.

Heavy-M does not improve KLD or CC. Its KLD is 0.2399, which is 6.26% worse than Light-M, and its CC is lower by 0.0022. Heavy-M improves SIM by only 0.0004, or approximately 0.05%. The per-image comparison is also close: Heavy-M performs better than Light-M on 47.08% of the images for KLD, 48.72% for CC, and 50.72% for SIM.

The difference in computational cost is much larger. Heavy-M has $6.12\times$ more parameters and $5.86\times$ more MACs than Light-M. Its median CPU latency is 55.160 ms, compared with 22.083 ms for Light-M, making it approximately $2.50\times$ slower. In this experiment, the larger encoder therefore adds a large computational cost without giving a clear improvement in the final accuracy.

Fig. 2 shows the same comparison using CC and median CPU latency. Light-M gives the most useful trade-off among the tested models. It improves the lightweight baseline without moving toward the much higher cost of Heavy-M. These

TABLE I

ACCURACY AND EFFICIENCY ON THE FROZEN 2,500-IMAGE INTERNAL TEST SET AT 169×256 . CPU LATENCY IS THE MEDIAN MODEL-ONLY FORWARD TIME WITH BATCH SIZE ONE AND ONE CPU THREAD. BOLD VALUES INDICATE THE BEST ACCURACY RESULT FOR EACH METRIC.

Model	KLD ↓	CC ↑	SIM ↑	Params (M) ↓	MACs (G) ↓	CPU (ms) ↓
MeanMap	0.7202	0.5570	0.5528	–	–	–
Light-S	0.2371	0.8735	0.7690	1.826	0.282	22.859
Light-M	0.2257	0.8813	0.7733	1.831	0.286	22.083
Heavy-M	0.2399	0.8792	0.7737	11.211	1.673	55.160

TABLE II
RESULTS ON THE 625-IMAGE OFF-CENTRE SUBSET.

Model	KLD ↓	CC ↑	SIM ↑
MeanMap	1.0073	0.4314	0.4585
Light-S	0.2743	0.8766	0.7504
Light-M	0.2608	0.8865	0.7551
Heavy-M	0.2813	0.8809	0.7529

results suggest that adding compact multi-scale fusion is more useful in this case than only increasing the size of the encoder.

E. Centre-Bias Stress Test

SALICON has a strong centre prior, so a fixed central map can still obtain reasonable scores without using the image content. To study this problem, the four methods were also evaluated on the 625 test images whose ground-truth centres of mass were farthest from the image centre. The results are reported in Table II.

MeanMap becomes much worse when the target attention is located away from the centre. Its KLD increases by 39.87%, while CC and SIM decrease by 22.55% and 17.06%. This is expected because MeanMap always predicts the same central distribution, regardless of the image.

The neural models are more stable on this subset. Their KLD increases by around 15.5–17.3%, while SIM decreases by only 2.4–2.7%. Their CC values remain stable and even improve slightly. This does not necessarily mean that the off-centre images are easier. A model can correctly locate an off-centre region and obtain a high CC, while still predicting a distribution that is too wide. This remaining difference is penalized more by KLD and SIM.

Light-M obtains the best off-centre result for all three metrics. Compared with Light-S, it reduces KLD by 0.0135, increases CC by 0.0099, and increases SIM by 0.0047. Heavy-M is also slightly worse than Light-M for all three metrics. This indicates that the multi-scale features help the model follow image-dependent attention when the centre prior is less useful.

F. Qualitative Analysis

Fig. 3 shows five cases selected using fixed criteria: the most central target, the most off-centre target, a sample near the median Light-M CC, the worst Light-M result, and the largest Light-M CC improvement over Light-S. This gives a more balanced view than showing only successful examples.

In the most off-centre example, sample 268435, the attended person is located near the lower-right part of the image. MeanMap does not capture this region, while all three neural models move their main prediction toward the person. Light-M obtains the highest CC for this image, although its KLD is slightly worse than Light-S. This example shows why the metrics can sometimes disagree.

The example with the largest Light-M improvement contains several separated attention regions. Light-S produces a simpler map and misses part of the target structure. Light-M distributes its prediction across more relevant regions, showing one situation where the shallower features provide useful spatial information.

The final example, sample 488346, is the worst Light-M failure. Its target contains several separate regions around a person and nearby objects. Light-M misses part of this structure and performs worse than Light-S, while Heavy-M gives the best prediction for this image. Multi-scale fusion therefore improves the average results, but it does not improve every individual example.

G. Limitations

The experiments use one deterministic SALICON split and one training run with seed 42 for each model. Since the annotations of the official SALICON test set are hidden, the reported scores are measured on an internal test split created from the official validation set. CPU latency also depends on the available hardware and implementation, and the reported values do not include image loading or preprocessing. The results should therefore be read as a controlled comparison between the tested models, rather than as a general ranking of all visual-saliency methods.

VII. CONCLUSIONS

The experiments show that using features from different encoder depths was more useful than simply increasing the size of the backbone. Light-M improved KLD, CC, and SIM over Light-S while adding only 0.27% parameters and about 1.20% MACs. Heavy-M was $6.12\times$ larger and $2.50\times$ slower than Light-M, but it did not improve KLD or CC. The off-centre evaluation also showed that the learned models were able to follow the image content better than the fixed MeanMap. For these reasons, Light-M provided the best balance between prediction quality and computational cost in this project.

During the project, I learned that a larger network does not always produce better results and that accuracy metrics and

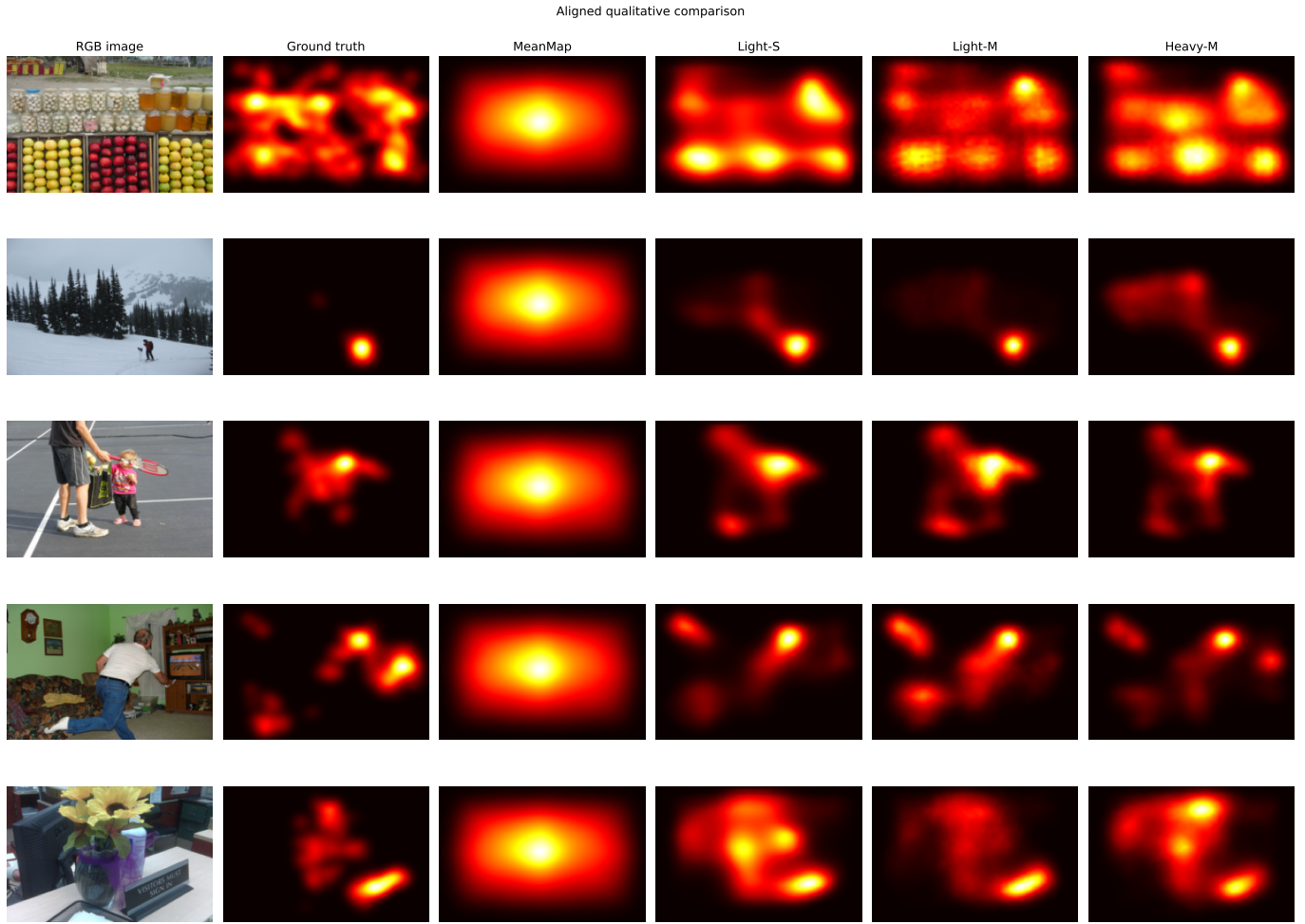


Fig. 3. Qualitative comparison on five selected internal-test images. Columns show the RGB image, ground truth, MeanMap, Light-S, Light-M, and Heavy-M. From top to bottom, the rows show the most central target, the most off-centre target, a sample near the median Light-M CC, the worst Light-M result, and the largest Light-M CC improvement over Light-S.

efficiency should be considered together. I also learned the importance of keeping the same preprocessing and evaluation conditions for every model. The main difficulty was discovering that the first Heavy-M evaluation used a different spatial resolution. This required retraining the model and repeating the comparison at 169×256 . The study is still limited to one split and one training seed. Future work could repeat the experiments with more seeds and eye-tracking datasets, and test techniques such as pruning, or knowledge distillation.

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